Syam Prakash

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SUMMARY

Systems Integration and Reliability Engineer with 3+ years at Tesla delivering automated end-of-line test systems for audio and radio hardware across 5+ vehicle programs. Specialized in systems-level reliability, cross-functional problem-solving, Python-based test automation, firmware validation, DSP algorithm testing, and production validation ensuring complex hardware performs in real-world conditions. Reduced EOL test cycle time by 60% and improved first-pass yield by building scalable CI/CD-integrated test infrastructure used daily by hardware, firmware, and factory teams.

TECHNICAL SKILLS

Languages	Python, C, C++, Shell Scripting, Groovy, Splunk SPL
Tools & Frameworks	Docker, Jenkins, GitHub Actions, MySQL, Dash, Plotly, Git
Hardware & Protocols	I2C, I2S, TCP/IP, A2B, CAN; VNA, Oscilloscope, Logic Analyzer, Spectrum Analyzer
Systems & Methods	Root Cause Analysis (RCA), test planning & specification, requirements traceability, telemetry & data analysis, hardware-firmware integration
Environments	Linux, Ansys HFSS, Keysight ADS, MATLAB, AWS

EXPERIENCE

Tesla — Senior Audio Test / Integration Engineer **Feb 2026 – Present**

Palo Alto, CA

- Own end-to-end system reliability across hardware, firmware, and algorithm teams; perform root cause analysis on multi-subsystem failures and drive corrective actions across electrical, firmware, and algorithm stakeholders.
- Design and maintain End of Line (EOL) and General Assembly test systems to detect faulty audio hardware, ensuring customers receive best-quality audio hardware in production vehicles.
- Conduct root cause analysis of audio and radio system failures; develop hardware fixtures for regression testing and Hardware-in-the-Loop (HIL) validation to improve fault detection coverage.
- Develop audio diagnostic and debugging tools for vehicles used by the audio team and customers to streamline troubleshooting and improve field support efficiency.
- Analyze factory production data to identify yield trends and optimize test sequences, improving first-pass yield and fault detection rates across vehicle programs.
- Develop Acoustic Buzz, Squeak, and Rattle (BSR) detection tests and perform subjective acoustic BSR analysis to characterize and resolve in-vehicle noise quality issues.
- Conduct hardware validation of audio amplifiers across multiple hardware revisions and production conditions, verifying electrical and acoustic performance against design specifications and ensuring field reliability.

Tesla — Software Test Engineer **Sep 2023 – Feb 2026**

Palo Alto, CA

- Architected and owned end-to-end EOL test infrastructure across 5+ vehicle programs, reducing cycle time by 60% and improving first-pass yield; defined test plans, managed requirements traceability, and drove cross-functional integration between hardware, firmware, and algorithm teams.
- Validated changes to audio hardware, DSP algorithms, and software for factory test impact; performed vehicle bring-up and regression testing to catch integration failures before production release.
- Developed real-time factory health dashboards using Dash and Plotly tracking KPIs across production lines; maintained CI/CD pipelines (Jenkins, GitHub Actions), Docker workflows, and MySQL test data stores.

- Triaged and prioritized bug reports, providing firmware developers with relevant logs and actionable details extracted from Splunk; reduced time-to-resolution for recurring audio defect classes.

Tesla — Radio Systems Intern **Nov 2022 – May 2023**

Palo Alto, CA

- Automated radio tuner validation pipeline, eliminating manual test steps and reducing qualification errors; built a Head Unit simulation tool enabling early bug detection before hardware availability.
- Validated in-car audio software; performed in-rush current, temperature cycle, and amplifier reliability tests; analyzed I2C and I2S signals to debug and document hardware issues.

Tracxn Technologies — Automation Engineer **Jul 2020 – Jul 2021**

Bangalore, India

- Upgraded Elasticsearch v6.9 to v7.9 enabling partial indexing and improved security; refactored Kafka consumer servers saving ~$40/day in infrastructure costs; managed team AWS infrastructure.

EDUCATION

M.S. Electrical & Electronics Engineering — Arizona State University **Aug 2021 – Aug 2024**

GPA: 3.92/4.00

B.Tech Electronics & Communication Engineering — Amrita Vishwa Vidyapeetham **Jul 2015 – May 2019**

GPA: 7.65/10